

Introduction to Machine Learning

Week 8

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1. (1 Mark) In Bagging technique, the reduction of variance is maximum if:

- (a) The correlation between the classifiers is minimum
- (b) Does not depend on the correlation between the classifiers
- (c) Similar features are used in all classifiers
- (d) The number of classifiers in the ensemble is minimized

Soln. A - This ensures diverse predictions that effectively average out errors

2. (1 Mark) If using squared error loss in gradient boosting for a regression problem, what does the gradient correspond to?

- (a) The absolute error
- (b) The log-likelihood
- (c) The residual error
- (d) The exponential loss

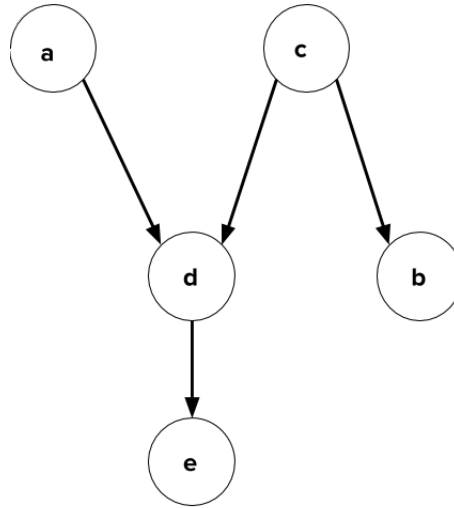
Soln. C - $\Delta(y - f(x; w))^2 = 2(y - f(x; w))\Delta f(x; w)$

3. (1 Mark) In a random forest, if T (number of features considered at each split) is set equal to P (total number of features), how does this compare to standard bagging with decision trees?

- (a) It's exactly the same as standard bagging
- (b) It will always perform better than standard bagging
- (c) It will always perform worse than standard bagging
- (d) Can not be determined

Soln. A - Random Forests differ from standard bagging by sampling a sub-set of features at every node

4. (1 Mark) **Multiple Correct:** Consider the following graphical model, which of the following are true about the model? (multiple options may be correct)



- (a) d is independent of b when c is known
- (b) a is independent of c when e is known
- (c) a is independent of b when e is known
- (d) a is independent of b when c is known

Soln. A, D - Refer to the lectures

5. (1 Mark) Consider the Bayesian network given in the previous question. Let “a”, “b”, “c”, “d” and “e” denote the random variables shown in the network. Which of the following can be inferred from the network structure?

- (a) “a” causes “d”
- (b) “e” causes “d”
- (c) Both (a) and (b) are correct
- (d) None of the above

Soln. D - Node “d” is dependant on both “a” and “c” and “e” can not cause “d”

6. (2 Marks) A single box is randomly selected from a set of three. Two pens are then drawn from this container. These pens happen to be blue and green colored. What is the probability that the chosen box was Box A?

Box	Green	Blue	Yellow
A	3	2	1
B	2	1	2
C	4	2	3

- (a) 37/18
- (b) 15/56

- (c) $18/37$
 (d) $56/15$

Soln. C -

Probability of choosing one box $P(A) = P(B) = P(C) = 1/3$. Here the event (E) is choosing the green and blue balls from the random box. Therefore, $P(E | A) = \frac{{}^3C_1 * {}^2C_1}{{}^6C_2} = 6/15 = 2/5$

$$P(E | B) = \frac{{}^2C_1 * {}^1C_1}{{}^5C_2} = 2/10 = 1/5$$

$$P(E | C) = \frac{{}^4C_1 * {}^2C_1}{{}^9C_2} = \frac{8}{72/2} = 2/9$$

$$\begin{aligned} P(A | E) &= P(E | A) / [P(E | A) + P(E | B) + P(E | C)] \\ &= \frac{2/5}{(2/5) + (1/5) + (2/9)} \\ &= 18/37 \end{aligned}$$

7. (1 Mark) **State True or False:** The primary advantage of the tournament approach in multiclass classification is its effectiveness even when using weak classifiers.

- (a) True
 (b) False

Soln. B - Refer to the lectures

8. (1 Mark) A data scientist is using a Naive Bayes classifier to categorize emails as either “spam” or “not spam”. The features used for classification include:

- Number of recipients (To, Cc, Bcc)
- Presence of “spam” keywords (e.g., “URGENT”, “offer”, “free”)
- Time of day the email was sent
- Length of the email in words

Which of the following scenarios, if true, is most likely to violate the key assumptions of Naive Bayes and potentially impact its performance?

- (a) The length of the email follows a non-Gaussian distribution
 (b) The time of day is discretized into categories (morning, afternoon, evening, night)
 (c) The proportion of spam emails in the training data is lower than in real-world email traffic
 (d) There’s a strong correlation between the presence of the word “free” and the length of the email

Soln. D - This scenario violates the Naive Bayes assumption of feature independence, as it the features are dependent on each other.

9. Consider the two statements:

Statement 1: Bayesian Networks are inherently structured as Directed Acyclic Graphs (DAGs).

Statement 2: Each node in a bayesian network represents a random variable, and each edge represents conditional dependence.

Which of these are true?

- (a) Both the statements are True.
- (b) Statement 1 is true, and statement 2 is false.
- (c) Statement 1 is false, and statement 2 is true.
- (d) Both the statements are false.

Soln. A - Bayesian Networks are structured as DAGs and each node represents a random variable, with edges indicating conditional dependencies.